Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI Agent Cybersecurity

TGS Ref No: TGS-2025053228

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 3.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
2.0	1 January 2025	Legacy release — 'Certified IT Specialist Cybersecurity' (Pearson) 2-day lesson plan.	Dr Alfred Ang
3.0	29 August 2026	AI-agent rewrite: retitled to AI Agent Cybersecurity (TGS-2025053228); three topics mapped to LO1–LO3; ten OpenClaw/Hermes labs; schedule cites deck slide ranges; assessment is WA-SAQ + Case Study on Day 2, 4:00–6:00pm.	Dr Alfred Ang

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Course Information

Course Title	AI Agent Cybersecurity
WSQ Course Reference	TGS-2025053228
Skills Framework (TSC)	Cyber Security (ACC-ICT-5002-1.1)
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 16 hours (14 hours instruction + 2 hours assessment)
Daily Timing	9:00 am – 6:00 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on labs on OpenClaw and Hermes Agent with authorised/synthetic targets
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Establish organizational cybersecurity policies and procedures to protect information confidentiality and integrity across AI-agent tools, skills, memory, context and permissions.
- LO2: Ensure adherence to cybersecurity policies and procedures in response to evolving threats, using AI-agent monitoring, compliance checks and controlled validation.
- LO3: Continuously improve cybersecurity policies and procedures to enhance organizational defenses, using vulnerability intelligence, incident response and a multi-agent resilience cycle.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book. Tests the underpinning knowledge (K1, K2).
- Case Study (CS) — a scenario-based assessment with three application tasks mapped to LO1 (A1), LO2 (A2) and LO3 (A3), 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- The final assessment is conducted on Day 2 from 4:00 pm to 6:00 pm (WA first, then the Case Study).
- Each instrument is graded Competent (C) or Not Yet Competent (NYC).
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Establishing AI-Agent Security Policies & Controls

Time	Duration	Topic / Activity
9:00–9:30	30 min	Welcome, course introduction, ground rules and mandatory

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		AM digital attendance (Slides 3–19)
9:30–10:45	75 min	**Core Concepts — Securing AI Agents: CIA triad, control categories, trust boundary, prompt contracts, least privilege (Slides 20–74)**
10:45–11:00	15 min	Tea break
11:00–13:00	120 min	Core concepts continued, then Topic 1 — Developing Cybersecurity Policies. Hands-on: Lab 1: Agent Security Policy & Trust Boundaries; Lab 2: Phishing & IOC Triage with Regex Generation (Slides 79–98)
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Topic 1 hands-on: Lab 3: Identity, Secrets, Encryption & Crypto Validation; Lab 4: Network Segmentation & IP Policy (Slides 99–118)
15:30–15:45	15 min	Tea break
15:45–17:45	120 min	Topic 1 recap, then Topic 2 begins. Hands-on: Lab 5: Agent Compliance Monitoring & Alert Triage (Slides 125–134)
17:45–18:00	15 min	Day 1 recap, Q&A and PM digital attendance

Total scheduled contact time: 480 minutes (8 hours).

Day 2 — Monitoring, Response & Continuous Improvement (+ Assessment)

Time	Duration	Topic / Activity
9:00–9:15	15 min	Day 1 recap and mandatory AM digital attendance
9:15–10:45	90 min	Topic 2 — Monitoring, Compliance & Response. Hands-on: Lab 6: Endpoint Hardening & Controlled Validation (Slides 135–144)
10:45–11:00	15 min	Tea break
11:00–13:00	120 min	Topic 2 hands-on: Lab 7: PCAP & Log Investigation; Topic 3 begins: Lab 8: Vulnerability Intelligence & Authorized FauxBank Assessment (Slides 145–170)
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Topic 3 — Continuous Improvement. Hands-on: Lab 9: Incident Response, Forensics & DR with Human Approvals; Lab 10: Capstone — Multi-Agent Cyber-Resilience Improvement Cycle (Slides 171–190)
15:30–15:45	15 min	Tea break

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15:45–16:00	15 min	**Course recap and exam preparation; Briefing for Assessment; TRAQOM survey and Assessment digital attendance**
16:00–17:00	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book**
17:00–18:00	60 min	**Case Study (CS) — scenario tasks mapped to LO1/LO2/LO3, 1 hour, open book. PM digital attendance**

Total scheduled contact time: 480 minutes (8 hours) — instruction 345 min, assessment 135 min.

Topic & Lab Reference (aligned to the learning outcomes and deck slides)

Topic / Learning Outcome	Weighting	Labs	Deck slides
Topic 01: Developing Cybersecurity Policies with OpenClaw and Hermes Agents (LO1)	40%	Lab 1, Lab 2, Lab 3, Lab 4	75–120
Topic 02: AI Agent Monitoring, Compliance and Response to Emerging Threats (LO2)	30%	Lab 5, Lab 6, Lab 7	121–156
Topic 03: Continuous Cybersecurity Improvement with Autonomous AI Agents (LO3)	30%	Lab 8, Lab 9, Lab 10	157–192

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